



Antimicrobial Susceptibility to Natural Substances of *Campylobacter jejuni* and *Campylobacter coli* Isolated from Italian Poultry

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Abstract

Campylobacter spp. isolated from poultry are a major public health concern. Antimicrobial resistance is increasing in general, which is associated with limited treatment of diseases in animal husbandry and in human as well. The use of natural substances (enterocins and phytoadditives) seems to be a promising alternative, either in the prevention or treatment of the gastrointestinal tract infections. A total 28 broiler carcasses from Italian poultry were sampled and 39 *Campylobacter* strains were isolated. Twenty-four isolates were identified as *Campylobacter jejuni* and 15 strains were identified as *Campylobacter coli* by polymerase chain reaction. All *Campylobacter* strains were determined to be resistant to nalidixic acid, ciprofloxacin, tetracycline (E-test), cefotaxim, and ampicillin using the disk diffusion method. Interestingly, the study revealed multidrug resistance of the *Campylobacter* strains; however, the strains were susceptible to natural substances, such as enterocins, oregano, and sage. Eighty-five percent of the *Campylobacter* strains were susceptible to at least one enterocins, and 77% of the strains were susceptible to enterocin (Ent) 9296, 69% were susceptible to Ent EM41, and 69% were susceptible to Ent A(P). Sage inhibited the growth of 38% of *C. jejuni* strains and 53% *C. coli* strains, whereas oregano inhibited the growth of 50% of the *C. jejuni* strains and up to 67% *C. coli* strains. Owing to the One health concept, application of enterocins and herbal extracts is a more ecological approach for preventing bacterial contamination during production and processing of poultry, which also leads to the protection of human health.

Keywords: *Campylobacter* spp., poultry, foodborne pathogens, sensitivity, resistance, antibiotics, enterocins, phytoadditives

Introduction

CAMPYLOBACTER SPP. COLONIZE the gastrointestinal tract of many farm and wild animals mostly without pathological consequences for the host (Skarp *et al.*, 2016). However, a recent study (Awad *et al.*, 2018) suggests that broiler flocks colonized with *Campylobacter* spp. may have an increased incidence of intestinal inflammation, diarrhea, and so on. The most commonly reported species causing human campylobacteriosis are thermotolerant *Campylobacter jejuni* (subspecies *jejuni*) and *Campylobacter coli*. Broilers are considered a reservoir of thermophilic *Campylobacter* spp., and these bacteria are the most im-

portant foodborne pathogen, associated with handling and consumption of contaminated raw or undercooked broiler meat.

Campylobacter infections in human is generally self-limiting, but severe prolonged or systemic infections can occur in populations that may require antimicrobial therapy (WHO, 2020). Because the incidence of antibiotic resistance of *Campylobacter* is increasing, treatment or prevention of *campylobacter* infections by antibiotics is becoming less successful. Drug resistance in bacteria is an increasing problem due to the possible transmission of resistant bacteria between animals and human through direct or indirect contact (Garcia-Migura *et al.*, 2014; EFSA, 2021).

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Intervention at the primary production level to reduce campylobacters is clearly preferable. Elimination or prevention of contamination by *Campylobacter* spp. may be possible using natural substances, such as phytoadditives or probiotic bacteria with their bioactive substances—bacteriocins. Bacteriocins are proteinaceous compounds lethal to bacteria other than the producing strain. Among bacteriocins, group of enterocins was classified, which are produced by some strains of the genus *Enterococcus*, mainly *Enterococcus faecium* species (Mareková *et al.*, 2003, 2007; Franz *et al.*, 2007).

Beneficial effects of enterocins have been reported and monitored *in vitro* and *in vivo* in previous studies of our laboratory (Lauková *et al.*, 2015; Ščerbová and Lauková, 2016). Having in mind, increasing antibiotic resistance, it is necessary to find alternatives. Therefore, the purpose of this study was to test *in vitro* susceptibility to enterocins and herbal extracts of *Campylobacter* spp. using enterocins and herbal extracts for their potential utilization in connection with One health concept.

Materials and Methods

Identification of *Campylobacter* spp. and hemolysis test

Samples from neck and breast skin of 28 broiler carcasses were collected from 23 different slaughterhouses in the Italy. *Campylobacters* were isolated according to Microbiology of Food and Animal Feeding Stuffs—Horizontal Method for Detection and Enumeration of *Campylobacter* spp. (ISO 10272-1:2017). Broiler samples were enriched in Preston broth with Preston supplement (Oxoid Ltd., Basingstoke Hampshire, United Kingdom) for 24 h, and were streaked on Karmali medium (Oxoid). Plates were incubated microaerobically at 42°C for 48 h. After that typical *Campylobacter* spp. colonies were plated on blood agar medium (Merck, Darmstadt, Germany) to check their purity. Differentiation between the species *C. coli* and *C. jejuni* was performed according to polymerase chain reaction protocol (Denis *et al.*, 1999).

Antibiotic phenotype resistance profile

Campylobacter strains were tested for antibiotic resistance according to The Clinical and Laboratory Standards Institute (CLSI, 2016) M45 guidelines by using the disk diffusion test and antimicrobial gradient method (minimal inhibitory concentration [MIC] evaluator). The disks (Oxoid, United

Kingdom) containing—ciprofloxacin (CIP; 5 µg), gentamicin (CN), ampicillin (AMP, 10 µg), erythromycin (ERY), azithromycin (AZM, 15 µg), streptomycin (STR; 25 µg), cefotaxim (CTX), chloramphenicol (CHL), nalidixic acid (NA), and tetracycline (TET; 30 µg). Evaluation of disk diffusion method was performed according to the manufacturer's instruction and sizes of inhibitory zones were measured in millimeters.

Based on the results of the disk diffusion method, strains were further tested using E-test strips containing the most commonly tested antibiotics related to *Campylobacter* spp. The MIC of E-test strips was following: ERY and TET (0.015–256 µg/mL), CN (Oxoid) and STR (0.064–1024 µg/mL), CIP (0.002–32 µg/mL) and NA (0.016–256 µg/mL; BioMérieux, Marcy l'Etoile, France). Antibiotic disks and strips were placed on the inoculated Muller–Hinton agar medium enriched with defibrinated horse blood (SR 0050; 5%; Oxoid). In both methods, plates were incubated at 42°C for 24–32 h in microaerophilic atmosphere (CampyGen™ 2.5L; Oxoid). Strains were classified as susceptible, intermediate, or resistant to the antimicrobial according to the CLSI (2016).

C. jejuni CCM 6191 and *C. jejuni* ATCC 33560 were used as positive controls. Antimicrobial free agar plates were included as a control for obligatory strain growth.

Susceptibility of *Campylobacter* spp. to enterocins and herbal extracts

The susceptibility of *C. jejuni* and *C. coli* to partially purified enterocins (10 µL; isolated and characterized at our Laboratory of Animal Microbiology, Institute of Animal Physiology, Centre of Biosciences of the Slovak Academy of Sciences, Košice, Slovakia) was tested using the agar spot method (De Vuyst *et al.*, 1996) on Trypticase soy agar (1.5% and 0.7%; Becton and Dickinson). The enterocins studied are summarized in Table 1. Inhibitory activity of enterocin was reported as the highest dilution of enterocin that inhibited growth of the indicator strain expressed in arbitrary unit per milliliter of culture medium (AU/mL).

The most susceptible strain *E. faecium* EA5 (isolated from piglet) was used as positive control. Inhibition effects of herbal extracts such as oregano and sage against growth of *Campylobacter* strains was also measured. Testing was conducted using the method of De Vuyst *et al.* (1996), and the size of the inhibitory zones was expressed in mm. The plant extract of *Origanum vulgare* (55% ± 3% of carvacrol) and *Salvia officinalis* (24% of thujone, 18% of borneol, 15% of

TABLE 1. LIST OF ENTEROCINS USED FOR TEST

Enterocin	Producer	Source	Activity in AU/mL	References
Ent EM 41	<i>Enterococcus faecium</i> EM41	Ostrich feces	25,600	Lauková <i>et al.</i> (2012)
Ent M	<i>E. faecium</i> AL41 ^a	Environmental strain	6400	Mareková <i>et al.</i> (2007)
Ent 55	<i>E. faecium</i> EF55	Chicken feces	51,600	Strompfová and Lauková (2007)
Ent A(P)	<i>E. faecium</i> EK13 ^b	Animal waste	25,600	Mareková <i>et al.</i> (2003)
Ent 9296	<i>E. faecium</i> 9296	Silage	25,600	Marciňáková <i>et al.</i> (2008)
Ent 412	<i>E. faecium</i> 412	Horse feces	25,600	Lauková <i>et al.</i> (2008)
Ent 4231	<i>E. faecium</i> 4231	Ruminal strain	3200	Lauková <i>et al.</i> (1993)
Ent 131	<i>E. faecium</i> HČ 13/1	Beaver feces	25,600	Unpublished data

^aAL41 = CCM8558.

^bEK13 = 7419.

AU/mL, arbitrary unit per mL.

TABLE 2. ANTIMICROBIAL SUSCEPTIBILITY ($\mu\text{g/mL}$) OF CAMPYLOBACTERS TO SIX ANTIBIOTICS BY E-TEST

Antimicrobial	MIC range	MIC ₅₀	Susceptible	Resistance
<i>Campylobacter jejuni</i> (n = 24)				
Erythromycin	1 to > 128	1	24/100%	0%
Gentamicin	0.12 to 0.5	0.25	24/100%	0%
Nalidixic acid	1 to > 64	> 64	9/37.5%	15/62.50%
Streptomycin	0.5 to 2	1	24/100%	0%
Tetracycline	0.5 to > 64	> 64	2/8.6%	22/91.6%
Ciprofloxacin	0.12 to > 16	> 16	4/16.6%	20/83.4%
<i>Campylobacter coli</i> (n = 15)				
Erythromycin	1 to > 128	1	10/66.7%	5/33.3%
Gentamicin	0.25 to 0.5	0.5	15/100%	0%
Nalidixic acid	8 to 64	> 64	1/6.6%	14/93.40%
Streptomycin	1 to > 16	2	14/93.4%	1/6.6%
Tetracycline	> 64	> 64	0%	15/100%
Ciprofloxacin	0.25 to 16	8	1/6.6%	14/93.4%

Number of strains/expressed as a percentage; MIC₅₀—the value defines the lowest antibiotic concentration at which at least 50% of the isolates were inhibited (within a species).

MIC, minimal inhibitory concentration.

cincole; Calendula company, Nová Ľubovňa, Slovakia) were tested. Plant extracts (concentration MIC 5) were applied in a dose 5 μL based on previous study (Szaboová *et al.*, 2013). Incubation was carried out at 42°C for 24–32 h.

Results

Resistance of *Campylobacter* spp. to antibiotics

The 24 *C. jejuni* and the 15 *C. coli* were generally resistant at least to two types of tested antibiotics. The results of antibiotic resistance tested by E-test are presented in Table 2.

The highest resistance was recorded to TET (*C. jejuni* 91.6%, *C. coli* 100%). The MIC₅₀ (required to inhibit the growth of 50% of the isolates) for NA was >64 $\mu\text{g/mL}$ (*C. jejuni* and *C. coli*). Occurrence of antimicrobial resistance tested by the disk diffusion method is shown in Table 3; the resistance to CTX and AMP of tested strains was evaluated as the highest (87.5% in *C. jejuni*, 80% in *C. coli*; in case of AMP it was 71% in *C. jejuni*, and 73% for *C. coli*. Antimicrobial multidrug resistance (MDR; resistance to at the least three different types of antibiotics) was found in 36 strains. The total resistance comprising both groups (*C. jejuni* and *C. coli*) is presented in Figure 1.

TABLE 3. ANTIMICROBIAL SUSCEPTIBILITY OF CAMPYLOBACTERS TO ANTIMICROBIALS BY DISK DIFFUSION METHOD

	AZM	CTX	AMP	CHL
<i>Campylobacter jejuni</i> (n = 24)				
Susceptible	21/87.5%	0%	5/21%	24/100%
Intermediate	0%	3/12.5%	2/8%	0%
Resistant	3/12.5%	21/87.5%	17/71%	0%
<i>Campylobacter coli</i> (n = 15)				
Susceptible	12/80%	2/13%	3/20%	15/100%
Intermediate	0%	1/7%	1/7%	0%
Resistant	3/20%	12/80%	11/73%	0%

Number of strains/expressed as a percentage.

AMP, ampicillin; AZM, azithromycin; CHL, chloramfenicol; CTX, cefotaxim.

Susceptibility of strains to enterocins and herbal extracts

Eighty-five percent of tested campylobacters were susceptible to at least one enterocin (Tables 4 and 5). Growth of as many as 30 MDR strains was inhibited by enterocins with the inhibitory activity 100 AU/mL (Tables 4 and 5). *Campylobacter* spp. were the most susceptible to Ent 9296 (30 strains, 77%). Six strains were resistant to all enterocins (Tables 4 and 5). In contrast, two strains (*C. jejuni* 2868/2 and *C. coli* 2804/2) were susceptible to all tested enterocins (eight).

Moreover, sage inhibited the growth of 38% of *C. jejuni* strains (nine susceptible) and 53% *C. coli* strains (eight susceptible) as illustrated by the production of medium size of inhibitory zones. Greater inhibition of *Campylobacter* spp. was produced by oregano with 50% of 12 *C. jejuni* strains being inhibited by the spice; 10 strains were inhibited reaching medium size inhibitory zones (10–25 mm) and 2 strains with high inhibitory zones (Table 4). Up to 67% (10 strains) of *C. coli* were susceptible to oregano; 9 strains with medium size of zones and 1 strain with high size zone (higher than 25 mm; Table 5).

Discussion

Campylobacteriosis is the most commonly reported human gastrointestinal infection in the European union. Up to 80% of campylobacteriosis are caused by the consumption of contaminated poultry meat or milk (EFSA, 2021). Our results demonstrated the high resistance rates of *Campylobacter* to CIP, TET, NA using the E-test and to CTX and AMP using the disk test. In particular, the resistance to CIP is still increasing; the total resistance to CIP was 87%. One of the treatment options for campylobacteriosis is TET. In our study, the total resistance to this antibiotic was almost 100%. Increased antimicrobial resistance of *Campylobacter* spp. has been confirmed in pigs, poultry meat, broiler chicken, and so on. (Matthew-Belmar *et al.*, 2015; Skarp *et al.*, 2016; Natsos *et al.*, 2021).

In contrast, the low antimicrobial resistance to CN might be due to the infrequent use of this antibiotic in poultry. Looking at the resistance rates within species, it has not been

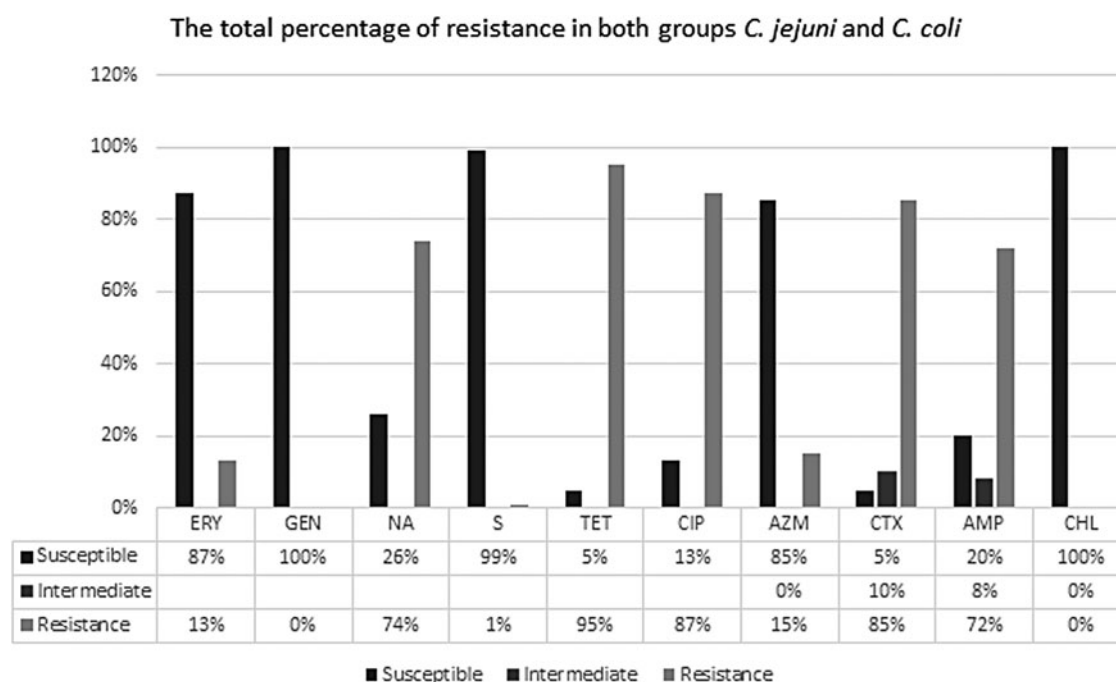


FIG. 1. The total antibiotic resistance of both groups together (group *Campylobacter jejuni* and group *Campylobacter coli*). AMP, ampicillin; AZM, azithromycin; CHL, chloramphenicol; CIP, ciprofloxacin; CTX, cefotaxim; ERY, erythromycin; Gen, gentamicin; NA, nalidixic acid; S, streptomycin; TET, tetracycline.

TABLE 4. SUSCEPTIBILITY TO ENTEROCINS AND PHYTOADDITIVES COMPARED WITH ANTIBIOTIC RESISTANCE IN *CAMPYLOBACTER JEJUNI*

Strain	EM 41	131	M	55	A(P)	9296	412	4231	Oregano	Sage	Resistance profile
<i>Campylobacter jejuni</i>											
Cj 2067/22	—	—	—	—	100	—	—	—	—	—	NA/TET/CIP/CTX/AMP
Cj 2583/22	100	—	—	—	—	100	100	100	—	—	NA/TET/CIP/CTX/AMP
Cj 2135/17	100	100	100	100	100	100	100	—	+	+	NA/TET/CIP/AZM/CTX
Cj 2136/22	100	—	100	—	100	100	—	100	+	+	NA/TET/CIP/CTX/AMP
Cj 2220/22A	100	—	—	100	100	100	100	—	++	+	NA/TET/CIP/CTX/AMP
Cj 2221/2	100	—	—	100	100	100	100	—	+	+	NA/TET/CIP/AZM/CTX/AMP
Cj 2401/2	—	—	—	—	—	100	—	—	++	+	NA/TET/CIP/CTX
Cj 2402/2	—	—	—	—	—	100	—	—	—	—	NA/TET/CIP/CTX/AMP
Cj 2411/2	100	—	—	100	100	—	100	—	—	—	TET/CTX
Cj 2412/2	100	—	—	100	100	100	100	—	—	—	TET/CTX/AMP
Cj 2461/2	100	100	100	—	100	100	—	—	—	—	CTX/AMP
Cj 2462/7	100	—	—	100	—	100	100	—	—	—	TET/CIP/CTX/AMP
Cj 2509/17	—	—	—	—	—	—	—	—	—	—	TET/CIP/CTX/AMP
Cj 2523/2	100	—	—	100	100	100	100	—	+	+	TET/CIP
Cj 2524/2	100	—	100	100	100	100	—	100	—	—	TET/CIP/CTX/AMP
Cj 2556/2A	—	100	100	100	100	100	100	100	+	—	TET/CIP/AZM/CTX/AMP
Cj 2557/12	100	—	—	100	100	100	100	—	+	—	NA/TET/CIP/CTX/AMP
Cj 2584/2A	—	—	—	—	—	—	—	—	+	—	TET/CTX/AMP
Cj 2589/2	100	100	—	100	100	100	100	—	+	+	NA/TET/CIP/AMP
Cj 2590/7	100	—	100	100	100	100	100	—	—	—	NA/TET/CIP/CTX
Cj 2597/7	100	100	—	—	—	100	—	100	+	+	NA/TET/CIP/CTX/AMP
Cj 2598/7	—	—	—	—	—	—	—	—	+	+	NA/TET/CIP/CTX/AMP
Cj 2804/17	100	—	—	100	100	100	100	—	—	—	NA/TET/CIP/CTX/AMP
Cj 2868/2	100	100	100	100	100	100	100	100	—	—	NA/TET/CIP

Enterocins: EM41, 131, M, 55, A(P), 9296, 412, 4231. The evaluation of the inhibitory zones for phytoadditives: the size of inhibitory zone <10 mm (—) no inhibitory zone; the inhibitory zone from 10 to 25 mm (+) medium size and the size of inhibitory zone higher than 25 mm (++) high size.

(—), none of isolates were susceptible; 100 (AU/mL), arbitrary units per mL; APM, ampicillin; AZM, azithromycin; CIP, ciprofloxacin; CTX, cefotaxim; ERY, erythromycin; NA, nalidixic acid; S, streptomycin; TET, tetracycline.

TABLE 5. SUSCEPTIBILITY TO ENTEROCINS AND PHYTOADDITIVES COMPARED WITH ANTIBIOTIC RESISTANCE IN *CAMPYLOBACTER COLI*

Strain	EM 41	131	M	55	A(P)	9296	412	4231	Oregano	Sage	Resistance profile
<i>Campylobacter coli</i>											
Cc 2067/22A	100	100	100	100	100	100	–	100	+	+	ERY/NA/TET/CIP/CTX/AMP
Cc 2220/2	100	–	–	100	100	100	100	–	+	+	NA/TET/CIP/CTX/AMP
Cc 2221/7A	–	100	–	–	100	100	100	100	–	–	NA/TET/CIP/CTX/AMP
Cc 2242/7	–	–	–	–	–	–	–	–	++	–	ERY/NA/TET/CIP/AZM/CTX
Cc 2319/7	100	–	–	–	100	–	–	–	–	–	NA/TET/CIP/CTX
Cc 2320/2	–	–	–	–	–	–	–	–	+	+	NA/TET/CIP/AZM/CTX/AMP
Cc 2402/7	100	–	–	100	–	100	–	100	+	+	NA/TET/CIP/CTX/AMP
Cc 2461/2A	100	–	–	100	100	100	100	–	+	+	ERY/NA/TET/CIP/CTX/AMP
Cc 2462/17	–	–	–	–	–	–	–	–	–	–	ERY/NA/TET/CIP/CTX
Cc 2556/2	100	–	–	100	100	100	100	–	+	+	NA/TET/CIP/CTX/AMP
Cc 2557/2	100	100	100	100	100	100	–	–	+	–	NA/TET/CIP/CTX AMP
Cc 2583/2	–	100	–	–	100	100	–	–	–	+	ERY/NA/TET/CIP/CTX
Cc 2584/2	100	–	–	100	100	100	100	–	+	+	TET/CTX/AMP
Cc 2804/2	100	100	100	100	100	100	100	100	–	–	NA/TET/CIP/AMP
Cc 2805/2	100	–	–	100	100	100	100	100	+	–	NA/S/TET/CIP/AZM/AMP

Enterocins: EM41, 131, M, 55, A(P), 9296, 412, 4231. The evaluation of the inhibitory zones for phytoadditives: the size of inhibitory zone <10 mm (–) no inhibitory zone; the inhibitory zone from 10 to 25 mm (+) medium size and the size of inhibitory zone higher than 25 mm (++) high size.

(–), none of isolates were susceptible; 100 (AU/mL), arbitrary units per mL; APM, ampicillin; AZM, azithromycin; CIP, ciprofloxacin; CTX, cefotaxim; ERY, erythromycin; NA, nalidixic acid; S, streptomycin; TET, tetracycline.

noted a significant difference in this study, except for ERY. *C. coli* strains displayed higher resistance to ERY (33%) compared with *C. jejuni* strains (0%). This is in agreement with the low resistance rate to ERY of *C. jejuni* reported by Matthew-Belmar *et al.* (2015). Thirty-six out of 39 of the *Campylobacter* spp. were determined to be multiresistant to antibiotics. This is extremely important finding from the standpoint of the spread of MDR pathogens to human through the contact between animals and human and the consumption of the contaminated animal products (Awad *et al.*, 2018).

According to EUCAST (2021) epidemiological cutoff values exceeding levels have been noticed for CIP (*C. jejuni* MIC₅₀ > 16, *C. coli* MIC₅₀ 8), NA and TET (MIC₅₀ > 64). Nowadays, many consumers try to live “healthier” in association with One health concept. Therefore, they look for healthy food products involving poultry. Owing to the high incidence of MDR strains worldwide, one of the options for the prevention or elimination of zoonotic infections in poultry production can be the use of enterocin-producing strains or their enterocins in poultry farming. Enterocins are small thermostable peptides with a broad antimicrobial spectrum against foodborne pathogens (Cintas *et al.*, 2000; Franz *et al.*, 2007).

In general, they have a broad spectrum against Gram-positive bacteria, but many of them are also active against Gram-negative bacteria (Line *et al.*, 2008; Ščerbová and Lauková, 2016). The bacteriocin structure and the sensitivity of target cells are crucial for the effect of the bacteriocin (Kjos *et al.*, 2009). *Campylobacter* spp. from Italian poultry were more susceptible to Ent 9296 than other enterocins used in this study. The growth of 33 *Campylobacter* spp. was inhibited by at least one enterocin; 30 out of those 33 campylobacters were multiresistant to antibiotics (antimicrobials). High efficiency of Ent 9296 against *Campylobacter* spp. tested was surprising and we would like to study it in more detail. It was repeatedly confirmed under *in vivo* and *in vitro* conditions that the antimicrobial activity of enterocins does not depend on the source of their producing strains.

In addition, enterocins and their producing strains can have other beneficial influences on various physiological parameters in animals (Lauková *et al.*, 2020; Zhang *et al.*, 2020). Another effective enterocins were Ent EM 41 and Ent A(P). Based on our previous results (Szabóva *et al.*, 2013), susceptibility to phytoadditives of the *Campylobacter* spp. was also tested.

Sage inhibitory effect against the growth of *Campylobacter* strains has not been previously reported. In general, phytoadditives have possessed antioxidant, anti-inflammatory, and antimicrobial effects. Effectivity of sage was manifested in 17 tested *Campylobacter* spp.; 16 out of which were MDR strains. A total 22 strains were susceptible to oregano; 21 out of which were MDR strains. Similar results were reported by van Alphen *et al.* (2012); they tested carvacrol, which is the main constituent of oregano. In future, an interesting study will be a test of the combined inhibitory effect of Ent 9296 and phytoadditives.

In conclusion, poultry *Campylobacter* spp. exhibiting multiantimicrobial/antibiotic resistance profile were susceptible to enterocins and herbal extracts with high inhibitory activity. Results from this study may provide a new approach to reduce *Campylobacter* spp. infection using preventive application of enterocins and/or plant extracts. In addition, this approach may provide a more ecological way to prevent infections by *Campylobacter* in livestock, especially having in mind reduction of resistant bacteria in environment.

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